TRUNG THANH TRAN

Applied AI Researcher & Builder · Foundation Models · Computer Vision · LLMs & Agentic Systems

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SUMMARY

AI researcher and engineering leader with 10+ years of experience building production AI systems across computer vision, search, speech, and LLM applications. Currently Head of AI at Pixta Vietnam (PIXTA Inc., Japan), leading systems that review 30K+ media assets daily and improve stock-media search conversion. Co-founder of two AI startups: face recognition (ClientScan, Oxford) and voice productivity (TakeNote.ai). Current focus: foundation models, efficient inference, LLM evaluation, and production AI systems for search, media review, and regulated domains.

EXPERIENCE

Pixta Vietnam (PIXTA Inc., Japan) 2019 – Present

Head of AI & Program Manager, PixtaStock (2026 – Present) · AI Expert → Senior AI Engineer (2019 – 2026)

- Lead AI research, innovation, and engineering management for the PixtaStock product — setting technical direction and overseeing delivery across all AI/ML projects (computer vision, NLP, search, image generation).
- Built and operate the automated media-review system — 30K+ photos, footage, and illustrations per day at >99% precision and ~70% recall — cutting manual-review cost by ~80% (≈ $100K–$150K per year).
- Shipped a learning-to-rank search system (LightGBM): +5% search conversion in production.
- Built an LLM-powered internal assistant saving ~$10K/year.
- Earlier: a Transformer auto-annotation model (+5% over the prior model) and a diversity-aware labelling method published at ICICT 2020.

Co-founder & AI Architect — ClientScan Ltd, Oxford, UK Jan 2023 – Present

- Co-founded a biometric-AI company delivering facial recognition and age verification for regulated UK retail, hospitality, and gaming (Challenge-25, gambling self-exclusion).
- Led and released **FaceAge ClientScan**, a deployable face-only age & gender model (DINOv3-ViT-L + CORAL): MAE 3.58 on LAGENDA 84K — outperforming the previous best specialised model, MiVOLO v2 (reported MAE 3.86), using face crops only — and avg MAE 4.90 across eight public age datasets.
- Architected the in-the-wild CCTV age-verification pipeline: face-quality gating, robust frame selection, and ByteTrack-based multi-frame aggregation for robustness to pose, lighting, and occlusion.

Co-founder — TakeNote.ai, Oxford, UK Jun 2023 – Present

- Co-founded a voice-AI product for meeting productivity: accurate transcription, summarization, and knowledge visualization.

Independent Research Collaborator — LLM Fairness & Cultural Bias Dec 2024 – Present

Part-time collaboration with Dr. Tarek Naous, Georgia Institute of Technology

- Co-author of CAMELLIA, a benchmark for cultural bias in LLMs across Asian languages; designed evaluation methods to surface and reduce where foundation models mishandle underrepresented cultures. Accepted to the EMNLP 2026 Main Conference.

AI Researcher — Fatima Fellowship Aug 2024 – Dec 2025

Fellowship research (part-time) with Dr. Ismet Dagli, Colorado School of Mines

- Developed efficiency and memory-optimization techniques for LLMs, vision-language models, and foundation models on resource-constrained and edge hardware.

AI Engineer / Consultant — Heligate JSC, Hanoi, Vietnam 2019 – 2020

- Built the first AI team; OCR document classification for a Japanese government client; a Rasa conversational system.

AI Scholar — Pi School, Rome, Italy 2018

- Conversational voice assistant for Sorgenia (energy utility), featured in Italian media.

Technical Leader & AI Engineer — Panasonic R&D Center, Hanoi, Vietnam 2015 – 2018

- Face recognition (ArcFace, F1 ≈ 0.95, 3D anti-spoofing); end-to-end ASR (DeepSpeech2); x-vector / LDA speaker ID and noise-robust wake-word models for smart speakers.

TECHNICAL SKILLS

Vision & Foundation Models: PyTorch, Vision Transformers (DINOv3), InsightFace, face & age estimation, CLIP & vision-language models

LLMs & Agentic Systems: fine-tuning (LoRA / QLoRA, DPO), RAG, LangGraph, Model Context Protocol (MCP), LLM evaluation & fairness

Search, Serving & Efficiency: learning-to-rank (LightGBM / LambdaRank), vLLM, NVIDIA Triton, ONNX, edge / distributed inference, quantization

Languages: Python, SQL · Vietnamese (native), English (professional)

OPEN MODELS & RESEARCH ARTIFACTS

FaceAge ClientScan — public face-only age & gender model release with PyTorch, ONNX, Safetensors, reproducible LAGENDA 84K benchmarking, and a full model card; MAE 3.58, CS@5 75.5%, gender accuracy 97.75%. Released on Hugging Face.

SELECTED PUBLICATIONS

CAMELLIA — LLM cultural-bias benchmark for Asian languages (with Georgia Tech, 2026) — **accepted to the EMNLP 2026 Main Conference**.

Three IGI Global book chapters on generative AI in education and ethics (2023–2024).

Diversity-aware image labelling (multi-model ensemble + Transformers) — ICICT 2020; earlier peer-reviewed GNSS / satellite-navigation research (2013–2014). Full list on Google Scholar.

AWARDS & HONORS

- United Nations / Italy Long-Term Fellowship, Politecnico di Torino (2013–2014); Summer Scholarship on GNSS, Institute of PNT of Japan, Tokyo (2014).
- Pi School of AI Scholarship + Co-runner-up, Notartel AI Hackathon, Rome (2018); first group in Asia to demonstrate Galileo-only positioning — featured in Inside GNSS and GPS World (2013).
- Runner-up, ICT Research Contest, HUST (2013); graduated ranked 1/120 (top of class).

EDUCATION

Politecnico di Torino — 2nd-Level Specializing Master, Navigation & Related Applications 2013 – 2014

Hanoi University of Science & Technology (HUST) — Engineer's Degree, Information Technology · Ranked 1/120 2008 – 2013

SERVICE & MENTORING

- Mentor, Vingroup 20,000 Applied AI Talent Program (AI20K); Startup Mentor & AI Advisor, VNU Innovation & Startup Incubator (VISI).
- Technical book reviewer, Manning Publications (AI, Deep Learning, LLMs, Agentic AI); admissions reviewer, MenaML Winter School 2026 (with Google DeepMind, QCRI, KAUST, Apple, InstaDeep).
- Judge, Vietnam AI Grand Challenge Hackathon (2019); mentor to master's students from Kyoto Institute of Technology & Ritsumeikan University.

REFERENCES

Available upon request.